Semester: 6th
Subject Name:-SPM & Code:- CS30012
Branch (s): - CSE, IT

SPRING **MID SEMESTER EXAMINATION-2025**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Software Project Management
[Subject Code: CS30012]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]

a) Which ones among the following can be considered as products:
I) User training
II) Trained user
III) Interviews completed
IV) Retrospective review
V) SRS document

b) What is the purpose of an activity network diagram?

c) Suppose that you are a software project manager, what approach you would prefer for effort estimation: bottom-up or top-down?

d) What is the significance of resource smoothing?

e) As a software project manager, how would you deal with 99% completion syndrome?

2. (a) "Motivation and application can often make up for shortfalls in innate skills, however, motivations vary from individual to individual." as a software project manager, what steps would you take to motivate your employees?

(b) Give a comparative analysis of project format vs functional format. As a software project manager, which organizational structure would you prefer? [2.5+2.5 Marks]

3. With scale factor values as following: PREC=2.48, FLEX=4.05, RESL=5.65, and remaining others to be extra high, and Post architecture effort multipliers as following: RCPX=2.72, PERS=2.12, FCIL=0.62 and remaining others to be nominal. If the project size is 25000 LOC and the average annual salary of the employees is 6 lakh, then what would be the estimated cost of the project? [5 Marks]

4. The same project if implemented in the US and in INDIA, gives different cash flow prediction as following, based on different discount rate and investment. As a project manager, where should you implement your project? [5 Marks]

		in INDIA	in USA
Discount Rate		7.50%	4.50%
Invested Capital		2500000	5000000
Cash Inflow -	Year 1	50000	1000000
	Year 2	100000	1500000
	Year 3	200000	2000000
	Year 4	300000	2000000
	Year 5	400000	1000000
	Year 6	500000	500000
	Year 7	1000000	300000
	Year 8	2000000	200000
	Year 9	4000000	50000

5. (a) A system has 4 numbers of inputs for each function types and all are of high complexity. If there is no external factors affecting the effort then, calculate the size using IFPUG function points.

(b) A system has 4 numbers of inputs for each function types. If there is no external factors affecting the effort then, calculate the size using Mark II function points.

(c) An embedded system has 4 numbers of inputs for each function types. Calculate the size using COSMIC function points, if there is no external factors affecting the effort.

[2+1.5+1.5 Marks]

*** Best of Luck ***